The Strategy — Two Signals, Three Positions

Geometric straightness ratio + Markov k=3 hidden states · SPY 2000-2025 · v5.0

Two Orthogonal Detectors — One Ensemble

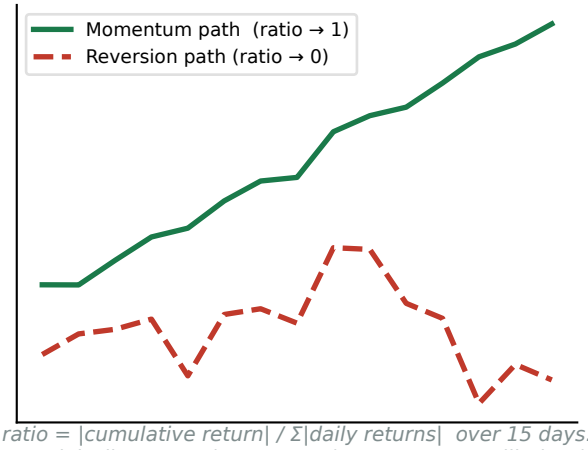
Geometric: path straightness over 15 days (no parameters to fit).
Markov: hidden 3-state model estimated from the full return history.
Combined as a simple mean — no weight fitting, no in-sample target.

Why It Works — Forward Return by Regime

Regime	Mean ret/day	T-stat (p)
Momentum	+0.018%	1.31 (0.189)
Mixed	+0.108%	3.21 (0.001)**
Reversion	-0.012%	-0.35 (0.729)

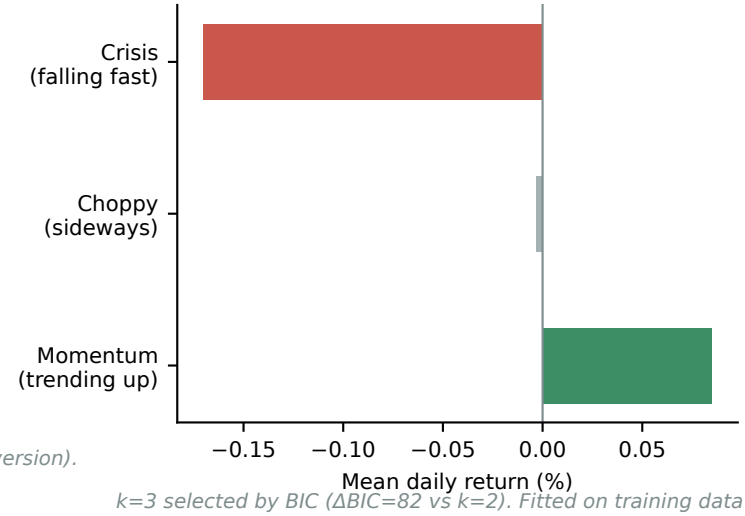
Mixed regime (detectors disagree) still drifts positive. T=3.21** is the strongest signal — justifies the half-position instead of defaulting to cash on uncertainty.

Detector 1 — Geometric (path shape, 15-day window)



ratio = |cumulative return| / Σ|daily returns| over 15 days.
Near 1.0 = straight-line move (momentum). Near 0.0 = oscillation (reversion).
Thresholds are adaptive percentiles — no fixed values to tune.

Detector 2 — Markov AR(1), k=3 hidden states



k=3 selected by BIC (ΔBIC=82 vs k=2). Fitted on training data only; Hamilton filter on test data — no look-ahead. Crisis (P>0.50) overrides all buy signals regardless of what the geometric detector says.

The Decision Rule — Full / Half / None

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score  =  mean( geometric_signal ,  markov_momentum_probability )
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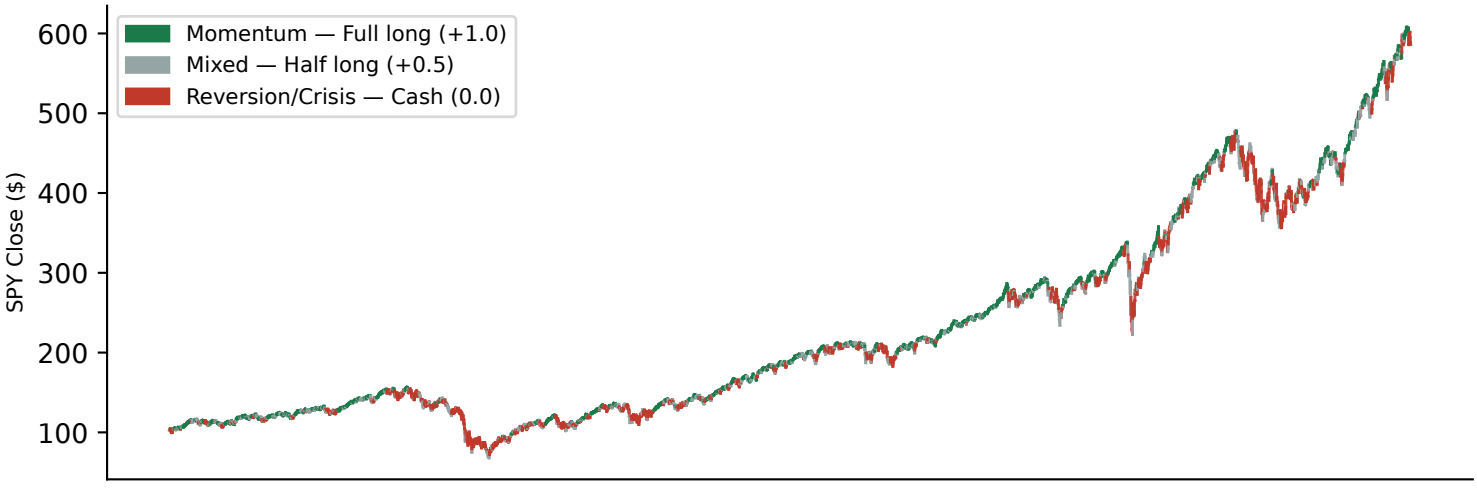
FULL LONG	HALF LONG	CASH
position = +1.0	position = +0.5	position = 0.0
score ≥ 0.65	0.35 ≤ score < 0.65	score < 0.35
Trend confirmed by both detectors.	*Detectors disagree — hold half.*	*Reversion or crisis — step aside.*
38% of trading days (2,017 days · SPY 2000-2025)	28% of trading days (1,488 days · SPY 2000-2025)	35% of trading days (1,857 days · SPY 2000-2025)

Models in Action — SPY 2000-2025 (Real Data)

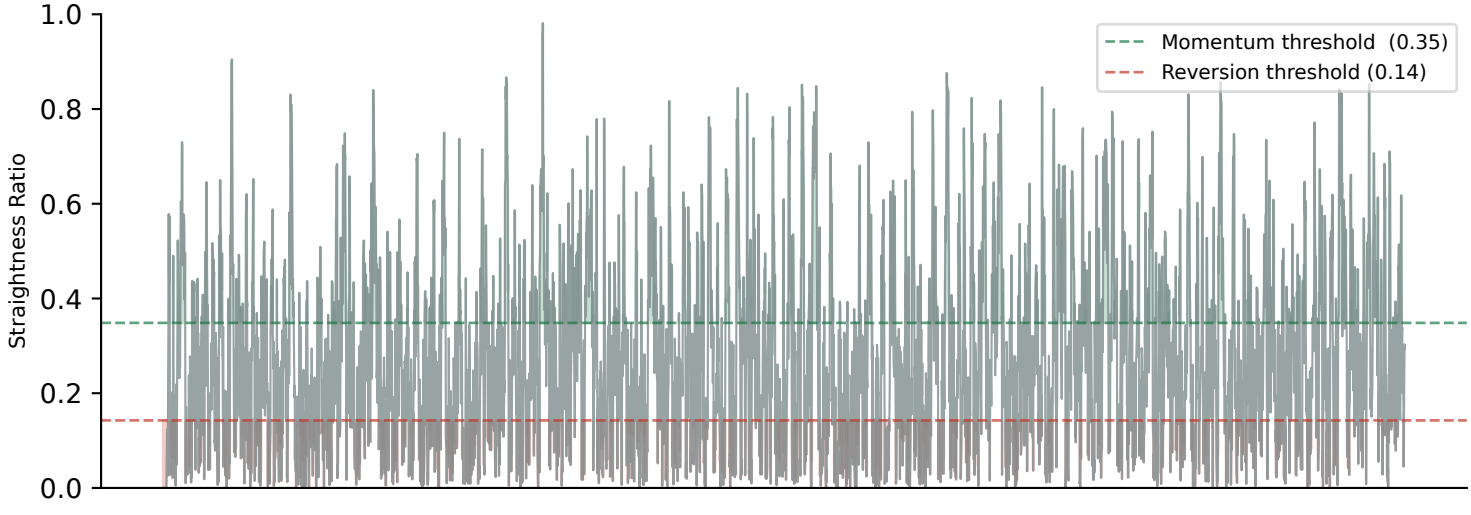
Price coloured by regime · geometric straightness ratio · Markov filtered probabilities

SPY Price — Coloured by Ensemble Regime



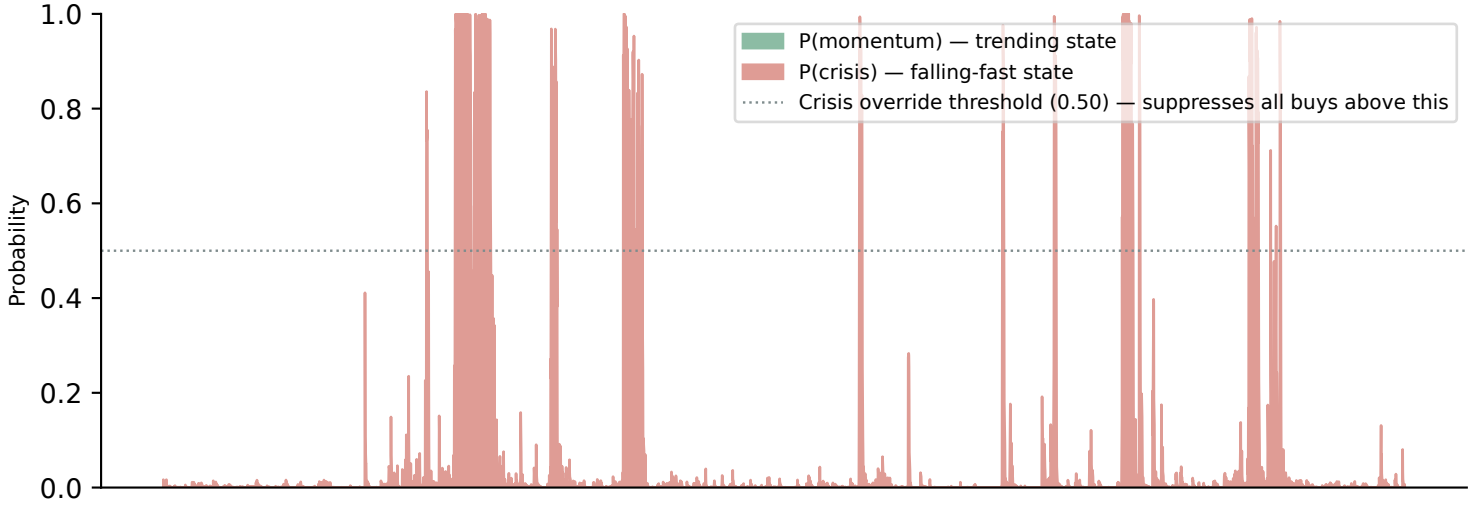
Green = momentum (+1.0 position). Grey = mixed (+0.5). Red = reversion or crisis (flat).
25 years: dot-com (2000-02), GFC (2008), COVID (2020), bear market (2022).

Geometric Detector — Straightness Ratio (15-day window)



Near 1.0 = price moved in a straight line (momentum). Near 0.0 = price zigzagged.
Thresholds are adaptive percentiles — no fixed values to hand-tune.

Markov Detector — Filtered State Probabilities (k=3)

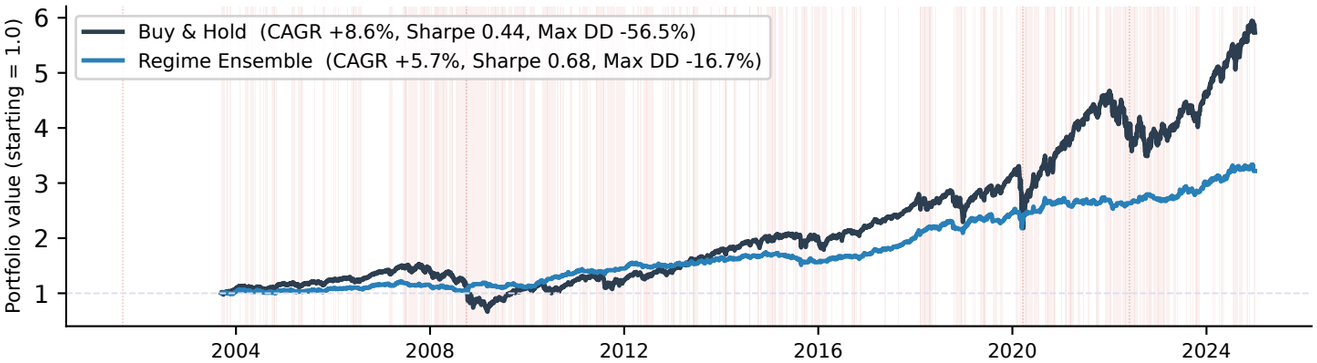


When P(crisis) > 0.50, buy signals are suppressed regardless of all other indicators.
Filtered probabilities only — Hamilton filter, no look-ahead bias.

The Edge — Performance vs Buy & Hold

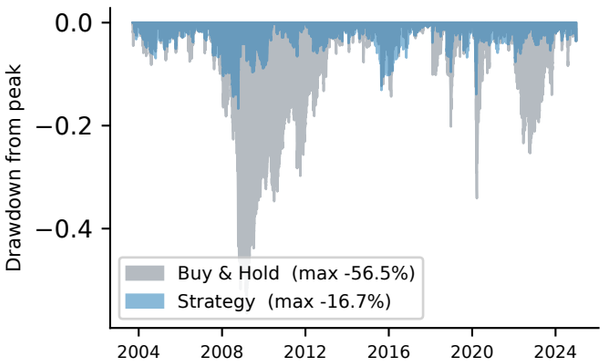
Risk-adjusted return, not raw return · SPY 2000-2025 · v5.0

Equity Curves — Strategy vs Buy & Hold (0 bps transaction costs)



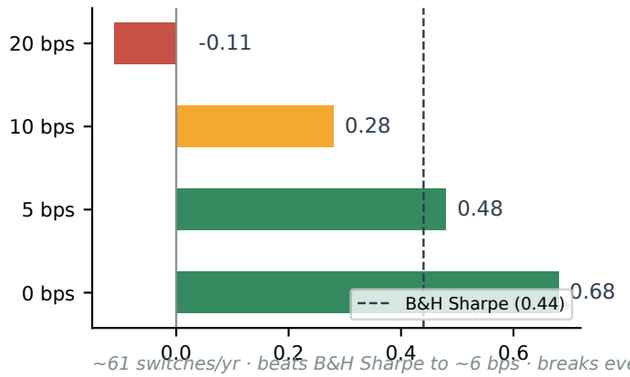
Red shading = periods the strategy held cash (reversion regime). Strategy sidesteps the worst of each major crash.
The edge is risk-adjusted: Sharpe 0.68 vs 0.44, drawdown -16.7% vs -56.5%. CAGR trails (+5.7% vs +8.6%).

Drawdown Comparison



Strategy trims drawdowns across all major crashes (dot-com, GFC, COVID, 2022).

Transaction Cost Sensitivity



~61 switches/yr · beats B&H Sharpe to ~6 bps · breaks even at ~17 bps.
--min-hold 3 reduces switches and extends the profitable range.

Honest Limitations — Read Before Drawing Conclusions

⚠ **Strategy CAGR trails buy-and-hold**
+5.7% vs +8.6% over 25 years at zero cost. The edge is Sharpe (0.68 vs 0.44) and max drawdown (-16.7% vs -56.5%). This is a risk-reduction strategy, not an alpha generator.

⚠ **Transaction costs are material**
~61 position switches/year. Sharpe beats B&H up to ~6 bps round-trip; strategy breaks even at ~17 bps. Use --min-hold 3 to cut switches.

⚠ **In-sample threshold calibration**
Percentile thresholds (70th/30th) and ensemble cutoffs (0.65/0.35) are fit on the full dataset. A real deployment requires expanding-window recalibration — see --expanding.

⚠ **Reversion signal is not statistically significant**
Regime T-stats: mixed=3.21** (p=0.001), momentum=1.33 (p=0.18), reversion=-0.35 (p=0.73). The strategy avoids reversion periods but does not short them.

⚠ **Calibrated on SPY — multi-asset not validated per-asset**
Thresholds are fitted on SPY. --multi-asset applies the same model to QQQ/IWM/TLT/GLD as a robustness check, but each asset's regime structure differs.